



Original Article

In-vitro antimicrobial susceptibility pattern of lipopeptide against drug resistant *Vibrio* species



Reem M. Aljowaie^a, Mohamed R. Abdel Gawwad^{b,*}, Dunia A. Al Farraj^a, Jose Kingsly H^{c,*}, P. Rajendran^d

^a Department of Botany and Microbiology, College of Sciences, King Saud University, P.O. Box 22452, Riyadh 11495, Saudi Arabia

^b Genetics & Bioengineering, Faculty of Engineering and Natural Sciences, International University of Sarajevo, Bosnia and Herzegovina

^c Vizhinjam Regional Centre of Central Marine Fisheries Research Institute, Vizhinjam, Thiruvananthapuram, Kerala, India

^d Kanyakumari Field Centre of Central Marine Fisheries Research Institute, Kanyakumari, India

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ABSTRACT

Background: The unrestricted application of antibiotics increased antimicrobial resistance in bacteria through horizontal gene transfer of resistant genes from the pathogenic sources and the evolution of multi-drug resistance organisms. The application of antibiotics caused severe risk to human health because animals may transmit diseases to humans. Hence, the search of novel antimicrobial agents from microbial sources is an urgent need.

Methods: A lipopeptide producing strain SU05 was isolated from the pond water by serial dilution method. The lipopeptide yield was improved after optimization method and the yield was analyzed using High Performance Liquid chromatography. The influence of wheat bran (0.5%–2.5%) and rice bran (0.5%–2.5%), pH (5.5–8.5), temperature (25–40 °C) were screened to improve the production of lipopeptides by strain SU05 in submerged fermentation. Antibacterial activity of crude lipopeptide was tested against *Vibrio anguillarum*, *Vibrio harveyi*, *Vibrio vulnificus*, *Vibrio salmonicida*, *Vibrio septicus*, *Vibrio fischeri*, and *Vibrio splendidus*. The influence of lipopeptide on enzymes and antimicrobial property was analyzed.

Results: Lipopeptide production was improved after nutrient supplements and optimization of physical factors. Lipopeptide showed potent activity against multi-drug resistant bacterial strains such as, *V. anguillarum*, *V. harveyi*, *V. vulnificus*, *V. salmonicida*, *V. septicus*, *V. fischeri*, and *V. splendidus*. Lipopeptide shows stability on various enzymes and this clearly revealed that the purified lipopeptide was highly stable in the presence of proteolytic enzymes. The findings suggest that lipopeptide SU05 characterized from the bacteria can survive at acidic environment in the intestine, and could be used to formulate fish feed.

Conclusions: The finding showed that the characterized lipopeptides synthesized by *B. amyloliquefaciens* SU05 had a broad spectrum antibiotic potential against multidrug resistant *Vibriosis* causing bacterial pathogens. They were highly stable at broad temperature and pH ranges. These results demonstrated stability of lipopeptide at extreme conditions. The stability and activity of lipopeptide at extreme climatic condition is also useful for the application in pharmaceutical and food processing industries.

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Introduction

Freshwater serves as an important water resource for irrigation, cooking and drinking. It is easily contaminated as a result of increased population, urbanization and associated with various dis-

eases in the developing and developed countries. The prevalence of various *Vibrio* species in the surface water is mainly influenced by location, season and based on analytical methods. Most of the *Vibrio* species are associated with various diseases in human and animal population. They are mainly linked to eruptions of *Vibrio* infections as a result of water and food contamination with sewage water or the exposure of wounds to animals or aquatic environment. Bacteria from the genus *Vibrio* comprise more than 100 species and most of the strains are pathogenic. These include, *Vibrio fluvial*, *V. tubiashi*, *Vibrio vulnificus*, *V. cholerae*, and *V. parahaemolyticus*

* Corresponding authors.

E-mail addresses: mrwab@ius.edu.ba (M.R. Abdel Gawwad), josekingslycmfri@gmail.com (J.K. H).

[1]. Recent analyses have revealed the role influenced by industrial and municipal wastewater and aquaculture as main driving force of various disease resistance genes in the aquatic environment. The unrestricted application of various commercial antibiotic drugs for disease management has resulted in evolution of ranges of multi-drug resistant bacterial pathogens in the natural environment. Chloramphenicol is widely used to control vibriosis in fish ponds. The application of chloramphenicol and other synthetic drugs in fish pond affects human health and increased drug resistance [2].

The beneficial bacteria, commonly called probiotics are widely used as alternate to chemically synthesized agents to improve the production of fish and to protect the natural environment [3]. Probiotic organisms improved the digestive enzyme activity of host species, inhibitory activity against bacterial pathogens, activation of immune defense, biodegradation of pond bottom soil containing organic sludge, improvement of water quality and promotion of animal growth and survival [4]. The important groups of probiotic organisms in aquaculture are yeast, bacteria, including *Bacillus*, and *Lactobacillus*, especially *Bacillus* species are widely used in aquaculture practice. Moreover, various problems have been reported frequently on probiotic organisms because of low effectiveness and high cost [5,6].

The microencapsulated forms of bacterial probiotics, microalgae and yeast were used to control vibriosis [7]. To improve the efficacy of the probiotic organism on the host species, analysis of the bacterial population in the host species is very important to reduce complication of secondary metabolite production and to improve efficacy. The pathogenic bacteria from the genus *Vibrio* involved pathogenesis in various organisms, including fishes. They are responsible for various diseases, and increased mass mortality in shrimp species [8,9]. Many commercially available antibiotics are used in aquaculture. Moreover, the application of antibiotics for prophylactic uses caused negative impact to the environment, fishes and humans. The substantial application of antibiotics induced drug resistance through horizontal gene transfer [10]. Probiotics have emerged as the eco-friendly alternatives to these chemically synthesized therapeutic molecules [11]. In order to select the suitable probiotics in aquaculture practice, the selected probiotics should be non-pathogenic and physiologically and biochemically well characterized. The selected probiotic organisms should be natural inhabitants of the host organism and able to grow and survive at gastrointestinal region. Also, they should maintain their activity and viability throughout the process condition [12]. Bacteria from the genus *Bacillus* was previously isolated from various sources, including marine fishes, bivalves and shrimp gut [13–15]. Administration of *Bacillus* species improved survival rate, and increased immune response in experimental animal, *P. monodon* [13]. In this study, *Bacillus amyloliquefaciens* isolated from the shrimp pond was subjected for the production of secondary metabolites and the probiotic potential was screened against various pathogens.

Materials and methods

Screening of antibacterial producing strains

The bacterial strain, *Bacillus amyloliquefaciens* SU05 was isolated from the shrimp pond water. The collected pond water was serially diluted and plated on nutrient agar plates. The plates were incubated for 24 h at $37 \pm 1^\circ\text{C}$ and the isolated bacterial strains were identified. The pathogenic bacterial strains such as, *Vibrio anguillarum*, *Vibrio harveyi*, *Vibrio vulnificus*, *Vibrio salmonicida*, *Vibrio septicus*, *Vibrio fischeri*, and *Vibrio splendidus* were collected from microbial type culture collections. They were grown in nutrient

broth medium and the bacteria strains were subcultured at $37 \pm 1^\circ\text{C}$. *Bacillus amyloliquefaciens* SU05 was cultured in lipoprotein production medium. The lipoprotein production medium consists of 1.5% wheat flour, 1.0% soybean cake powder and the pH of the medium was adjusted to 7.0 using 1 N HCl or NaOH.

Analysis of drug resistance pattern of *Vibrio* species

The selected seven bacterial strains from the genus *Vibrio* were cultivated aerobically in nutrient broth medium for 24 h at $37 \pm 1^\circ\text{C}$. They were seeded separately on agar plates and antibiotic was added and incubated for 24 h and the zone of inhibition (mm was observed).

Characterization of probiotic strain SU05 isolated from the pond water

The bacterial strain SU05 isolated from the fish pond was subjected for characterization studies. It was grown in nutrient broth medium and the growth performance was analyzed. The optimum growth, colony morphology and biochemical characterization was performed. The genomic DNA of the strain SU05 was isolated and further purified using Genomic DNA extraction and purification kit (Qiagen, Germany) based on manufacture instructions. The purity of the DNA was also checked using a Nanodrop machine. The forward and reverse primers were used for the amplification of 16S rDNA gene according to the methods suggested by Al-Dhabi et al. [16]. The primers used were 5'-CAAGTCGAGCGGACAGATG GGAGCT-3' (forward) and 5'-AGCT CCCATCTGTCC GCTCGACTTG-3' (reverse). The sequences were submitted in Genbank database and accession number was assigned.

Crude metabolite extraction

The strain SU05 was cultured in Erlenmeyer flask containing 100 mL minimal medium and incubated for 24 h and crude lipopeptide was extracted. The minimal medium composed of (g/L), ammonium chloride – 0.4 g, sodium chloride – 0.5, potassium phosphate – 3, disodium phosphate – 6.6 and pH 6.8 ± 0.2 . The culture was centrifuged at 13,000 rpm for 10 min and the cell free supernatant was aseptically filtered using 0.22 μm syringe filter and the filtrate was stored at 4°C for antibacterial assay [4].

Improved production of metabolites

The influence of wheat bran (0.5%–2.5%) and rice bran (0.5%–2.5%), pH (5.5–8.5), temperature (25–40 $^\circ\text{C}$) were screened to improve the production of lipopeptides by *B. amyloliquefaciens* SU05 in submerged fermentation. To the 250 mL Erlenmeyer flask, 100 mL minimal medium was prepared and 1% (v/v) inoculum was introduced and lipopeptide production was assayed.

Antibacterial assay

The crude sample was used for antibacterial assay. The pH of the supernatant was adjusted to 2.5–3.0 using 1 N Hydrochloric acid and the mixture was stored for 6 h at 4°C . The acid precipitate was further centrifuged at 12,000 rpm for 15 min at 4°C and the final pellet was washed with 0.1 N HCl solution. Then the pellet was neutralized with 1 N NaOH and freeze dried. The freeze dried sample was used for antibacterial analysis. The dried sample was extracted three times with methanol and the extract was further evaporated. Antibacterial activity of the crude lipopeptide was tested against *V. anguillarum*, *V. harveyi*, *V. vulnificus*, *V. salmonicida*, *V. septicus*, *V. fischeri*, and *V. splendidus*. The crude lipopeptide sample (25 μL)

Table 1

Antibiotic susceptibility of bacterial strains against multidrug resistant vibriosis causing bacterial strains.

Antibiotics	<i>V. anguillarum</i>	<i>V. harveyi</i>	<i>V. vulnificus</i>	<i>V. salmonicida</i>	<i>V. septicus</i>	<i>V. fischeri</i>	<i>V. splendidus</i>
Ampicillin	R	R	R	R	R	R	R
Chloroamphenicol	S	R	R	S	S	R	S
Tetracycline	R	R	R	R	R	R	R
Oxytetracycline	S	S	S	S	S	S	S
Kanamycin	S	S	S	R	S	S	S
Colistin	R	S	R	S	R	S	R
Neomycin	S	R	S	S	S	S	R
Levofloxacin	R	R	S	R	R	S	S
Streptomycin	S	S	R	S	S	R	S
Amoxicillin	S	R	S	S	R	R	S

R: resistance; S: sensitive.

was loaded on the sterile disc and incubated the plates for 24 h and zone of inhibition was measured (mm) [6].

High performance Liquid Chromatography (HPLC) analysis of lipopeptides

High Performance Liquid Chromatography was used for the determination of lipopeptides from the sample. The crude sample was loaded on C18 HPLC column and retained the purified lipopeptides. The applied cartridge was washed with Millipore water and the bound lipopeptides were carefully eluted using 100% methanol. The solvent was evaporated from the sample and crude sample was dissolved in absolute methanol (0.1 mL).

Growth kinetics of strain SU05

Growth kinetics of the bacterial strain was analyzed for 60 h. A loopful culture of SU05 was inoculated in the nutrient broth medium (Himedia, Mumbai, India). It was incubated at $37 \pm 2^\circ\text{C}$ for 60 h. After every 6 h interval, the growth was analyzed at 600 nm using a UV Visible spectrophotometer.

Protease assay

The cell free extract (0.1 mL) was mixed with 1.0 mL substrate (casein) prepared in Tris–HCl buffer (pH 7.5, 0.1 M). The reaction mixture was further incubated for 10 min at $37 \pm 2^\circ\text{C}$. After 10 min reaction, 10% TCA (trichloroacetic acid) (2.5 mL) was added and mixed to stop the enzymatic reaction. The sample was centrifuged and the precipitate was discarded. To the clear supernatant, sodium carbonate (2.0 mL, 500 mM), Folin–phenol reagent (0.5 mL) was added the tyrosine content released was measured at 630 nm using a UV–vis spectrophotometer [17]. One enzyme unit was described as the amount of the crude sample that liberated 1.0 μg equivalent of L-tyrosine/min under standard assay conditions.

Amylase activity

The cell free supernatant (0.1 mL) was mixed with 1.0 mL substrate (soluble starch) prepared in sodium phosphate buffer (pH 7.0, 0.1 M). The reaction mixture was then incubated at $37 \pm 2^\circ\text{C}$ for 10 min. After 10 min, DNS reagent was added to all test tubes and placed on a boiling water bath for 10 min. To the control, enzyme was incorporated after the addition of DNS reagent. One enzyme unit was defined as the amount of enzyme required to release 1.0 μM equivalent reducing sugar per minute under the standard assay conditions.

Table 2Antagonistic activity of lipopeptide produced by *B. amyloliquefaciens* SU05 against multi-drug resistant bacteria.

Pathogenic bacteria	Zone of inhibition (mm)
<i>V. anguillarum</i>	22 ± 2
<i>V. harveyi</i>	26 ± 1
<i>V. vulnificus</i>	19 ± 1
<i>V. salmonicida</i>	22 ± 2
<i>V. septicus</i>	20 ± 1
<i>V. fischeri</i>	18 ± 3
<i>V. splendidus</i>	23 ± 1

Effect of pH and temperature on antibacterial activity of lipopeptide

To analyze the influence of pH and temperature on lipopeptide activity, the purified lipopeptide samples were incubated at various pH ranges (4.0–10.0). The sample was incubated for 3 h at $4 \pm 1^\circ\text{C}$. *V. harveyi* was used as the indicator organism, and the antibacterial activity of the sample was measured on Mueller Hinton Agar plates. Samples were neutralized before to perform antibacterial assay. To analyze temperature stability of the enzyme sample was incubated at various temperatures (40°C – 90°C) for 30 min. It was cooled down to ambient temperature and antibacterial activity was tested using indicator organism.

Enzyme stability analysis

The lipopeptide sample was used for the determination of enzyme stability analysis. Briefly, the tested enzymes such as, bromelain, chymotrypsin and trypsin were diluted appropriately in Tris–buffer (pH 8.0, 0.05 M) and pepsin was diluted in glycine–HCl buffer (pH 2.0, 0.05 M). Enzyme reaction was performed at 0.5 mg/mL concentration and incubated for 3 h. The final residual antibacterial activity was determined using antibacterial assay by indicator strain.

Results

Characterization of multidrug resistant fish pathogenic bacteria

In order to determine multi-drug resistant ability to the commercial antibiotics, antibacterial sensitivity test was performed. The findings in Table 1 showed that almost all selected *Vibrio* spp. were highly resistant to most of the selected antibiotics, suggesting that all of them were multidrug-resistant isolates.

Antagonistic activity of lipopeptide against multi-drug resistant bacteria

Lipopeptide was used to assay antibacterial activity against all selected *Vibrio* spp. and the result was tabulated in Table 2.

Table 3

Production of lipopeptide by *B. amyloliquefaciens* SU05 using wheat bran and rice bran medium at various concentrations.

Concentration	Zone of inhibition (mm)	
	Wheat bran	Rice bran
0.5	17 ± 2	9 ± 1
1	18 ± 1	15 ± 2
1.5	24 ± 2	20 ± 2
2	20 ± 1	18 ± 3
2.5	16 ± 2	14 ± 1

The commonly used antibiotics affected environment and human health, hence alternative methods are required to solve the environmental impacts. Multi-drug resistance pattern of the selected *Vibrio* spp. indicated increased demand of novel antibiotics from natural sources. The bacterial isolates that were screened from the fish pond water were analyzed in order to test potential against the multi drug resistant *Vibrio* strains involved in fish spoilage. The present finding revealed that the strain SU05 showed broad spectrum antibacterial activity against almost all tested *Vibrio* species. Hence, the bacterial isolate SU05 was applied for further investigations.

Characterization of *B. amyloliquefaciens* SU05

The bacterial colonies were flat, irregular in shape, and white in colour. It has the ability to hydrolyze casein and starch. It was able to grow at various temperatures (20–40 °C) and broad pH (4.0–8.0) ranges. It was rod shaped and survived up to 5% sodium chloride concentration. Based on morphological, biochemical and molecular methods identification, the selected bacterial strain was identified as *B. amyloliquefaciens* SU05. BLAST analysis revealed that the selected bacterial strain SU05 was closely related with other *B. amyloliquefaciens* strains. The sequence similarity was 98% and it was concluded that the strain SU05 was belonged to *B. amyloliquefaciens* strains.

Enhanced production of bioactive molecules by *B. amyloliquefaciens* SU05 against multidrug resistant *Vibrio* sp

The selected bacterial strain *B. amyloliquefaciens* SU05 could produce antibacterial metabolites in 1.5% wheat bran in the culture medium and at higher concentrations bioactive molecule production declined marginally. In the case of rice bran medium, 2% substrate rice bran induced the production of secondary metabolites and moderately declined at 2.5% concentration (Table 3). The optimum temperature for the production of lipopeptide was 7.5 and production was high at 40 °C (Table 4).

Table 5

Growth kinetics, biomass production, enzyme assay and antibacterial activity of metabolite produced by *B. amyloliquefaciens* SU05.

Incubation time (h)	Biomass (g/L)	Protease (U/mL)	Amylase (U/mL)	Zone of inhibition (mm)
6	0.18 ± 0.31	3.8 ± 0.32	1.2 ± 0.3	ND
12	0.42 ± 0.16	10.7 ± 0.56	2.9 ± 0.42	ND
18	1.28 ± 0.32	49.2 ± 2.4	5.2 ± 0.2	10 ± 2
24	1.97 ± 0.24	68.3 ± 1.1	19.7 ± 1.4	18 ± 1
30	2.2 ± 0.17	85.3 ± 2.8	40.3 ± 2.5	24 ± 2
36	2.16 ± 0.35	78.4 ± 3.3	50.3 ± 1.1	20 ± 2
42	1.99 ± 0.14	75.2 ± 1.4	45.3 ± 2.2	17 ± 1
48	1.842 ± 0.18	70.3 ± 3.3	39.1 ± 1.7	15 ± 2
54	1.74 ± 0.52	60.4 ± 1.2	30.7 ± 1.3	15 ± 1
60	1.549 ± 0.31	59.2 ± 3.5	20.1 ± 1.9	13 ± 2

Table 4

Effect of temperature and pH on lipopeptide production by *B. amyloliquefaciens* SU05.

Temperature (°C)	Zone of inhibition (mm)	pH	Zone of inhibition (mm)
25	18 ± 2	5	20 ± 2
30	20 ± 1	6	20 ± 1
35	22 ± 1	7	23 ± 2
40	26 ± 2	8	26 ± 1
45	24 ± 1	9	22 ± 2
50	24 ± 1	10	20 ± 3

Growth kinetics, protease and amylase activity

The growth kinetics of *B. amyloliquefaciens* SU05 was depicted in Table 4. The optical density of the strain was 0.25 ± 0.06 after 6 h incubation and it reached maximum after 30 h (2.2 ± 0.17 OD at 600 nm). Protease production was growth associated and maximum protease production was observed after 30 h incubation (85.3 ± 2.8 U/mL), and amylase production also found to be maximum at 36 h incubation (50.3 ± 1.1 U/mL). Antibacterial activity was found to be high after 36 h incubation and the zone of inhibition was 24 ± 2 mm (Table 5).

Effect of pH, temperature and proteases on lipopeptide stability

The effect of pH and temperature on lipopeptide stability was analyzed. The stability of lipopeptide in the presence of proteolytic enzymes, physical conditions such as temperature, and pH were analyzed. Hence, to analyze the stability, the purified lipopeptide was subjected to various pH values and the antibacterial activity was checked against indicator organism. The lipopeptide extracted from *B. amyloliquefaciens* SU05 showed stability after 3 h incubation at 37 °C and no significant difference on antibacterial activity (Table 6). Lipopeptide showed stability on various enzymes and this clearly revealed that the lipopeptide was stable in the presence of proteolytic enzymes. The findings suggest that lipopeptide SU05 characterized from the bacteria can survive at acidic environment in the intestine, and could be used to formulate fish feed.

Discussion

Overuse and misuse of various commercially available antimicrobial agents induced drug-resistant among animals and affect the environment. The application of single antibacterial agent may induce cross-resistance to various antimicrobial agents used to cure human and animal diseases. The continuous application of chlortetracycline was associated with tetracycline and ampicillin resistance in the fecal coliforms isolated from the swine farms [18]. Application of sulfamethazine and chlortetracycline in ruminants diet induced the evolution of tetracycline- and ampicillin-resistant *E. coli* [19]. The unrestricted application of antibiotics caused severe risk to human health because animals may transmit diseases to

Table 6

Antibacterial activity of lipopeptide at various pHs, temperatures and proteolytic enzymes.

Physical factors and enzymes		Zone of inhibition (mm)
pH	6	18 ± 1
	7	21 ± 2
	8	24 ± 1
	9	20 ± 3
	10	14 ± 1
Temperature (°C)	50	24 ± 2
	60	24 ± 2
	70	23 ± 2
	80	21 ± 3
	90	21 ± 1
	100	21 ± 2
Enzymes	Pepsin	22 ± 1
	Trypsin	24 ± 1
	Bromelain	19 ± 3
	Chymotrypsin	17 ± 1

humans. *Vibrio* is one of the pathogenic organisms involved in mass mortality in hatcheries and shrimp farms. It has been reported the pathogenic properties of multi-drug resistant *Vibrio* species such as, *V. vulnificus*, *V. salmonicida*, *Vibrio ordalii* and *V. anguillarum* isolated from the environment. Increased antibiotic resistance in the aquatic eco-system results spontaneous mutation by available antibiotics in the particular environment by horizontal gene transfer and its effective stimulation between various genera and species including, fish pathogens, marine bacteria and human pathogenic bacteria. The most important concern is emergence of new strains that are highly resistant to many antibiotic agents. Application of antibiotics in the aquaculture practice effectively reaches wild shellfish and finfish in and around the aquaculture sites and is used for consumption. Moreover, the use of commercial antibiotics is used to control diseases in animals. In order to test the novel antibiotics from *Bacillus* species from the shrimp farm water, the antimicrobial potential has been tested against vibriosis causing multi-drug resistant bacteria [5–8]. The present finding indicated that the antibacterial agent produced by the strain has the potential to prevent the growth of various tested *Vibrio* species.

Bacillus species produce polypeptide antibiotic agents, such as, tyrotricidin, polymyxin, gramicidin S, and bacitracin, which are highly active against number of Gram-negative and Gram-positive bacteria [20,21]. In this study, *B. amyloliquefaciens* SU05 was capable of inhibiting the growth of various *Vibrio* spp. Previous studies revealed the importance of optimum environmental factors (temperature and pH) on lipopeptide production [22]. Bacteria from the genus *Bacillus* involved in the production of polypeptides which were effective against various bacterial species. Wheat bran and Rice bran medium were used as the carbon source for the production of lipopeptide and 1.5–2.0% was optimum for the biosynthesis of lipopeptide. The production of lipopeptide was growth associated and the selected *Bacillus* species produced considerable amount of protease and amylase after 36 h. The lipopeptide synthesized by *B. amyloliquefaciens* SU05 showed antibacterial activity against various bacterial pathogens. The enzyme activity varied widely among the bacterial strains. The purified lipopeptide showed maximum activity at alkaline pH, at 40 °C and retained antimicrobial activity after enzymatic treatment. The present finding revealed resistance of lipopeptides against amylases and proteases and temperature variations. The lipopeptide molecule could be cyclic peptide containing unique amino acids [23,24]. The present findings revealed that *B. amyloliquefaciens* SU05 could be survived and biosynthesized lipopeptide at broad temperature and pH ranges. The present study revealed the bioactive

role of bacteriocin-like molecules, which may critically contribute antibacterial activity [25,26].

Conclusions

The lipopeptide produced by the bacterium, *B. amyloliquefaciens* SU05 showed potent activity on *V. anguillarum*, *V. harveyi*, *V. vulnificus*, *V. salmonicida*, *V. septicus*, *V. fischeri*, and *V. splendidus*. Lipopeptide has various advantages because of high yield, simple protocol for downstream processing, simple to scale up and stability at extreme pH and temperature.

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Competing interests

None declared.

Ethical approval

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